

Name:

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Signature :

MS101 – Makerspace 2024-25/I (Autumn Semester)

Oct 23, 2024 (Wed)

EE Quiz

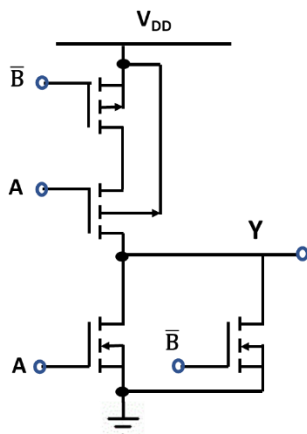
Time: 45 min (for Parts A and B)

Part A: EE Quiz, Marks 15

1. Part A: EE Quiz has 4 pages (pages 1 to 4) and 9 questions.
2. Part B: EE Project Quiz has 2 pages (pages 5 and 6) and 6 questions.
2. Write your **answers only in the space provided for answers**. Answers written at any other place will not be checked. You may use the page margins also for rough work.
3. No explanations / clarifications will be given to any of the questions.
4. No negative marks for wrong answers, **however steps are required for all numerical answers.**
5. **In case of questions with multiple-choices, credit will be given only if all correct options are identified.**

Q.no.	1	2	3	4	5	6	7	8	9		TOTAL
Marks											

1. Consider the given CMOS digital circuit:



Choose all the options which are correct:

- A) $Y = \overline{A + B}$
- B) $Y = \overline{\overline{A} + B}$
- C) $Y = \overline{A + \overline{B}}$
- D) $Y = \overline{A} \cdot B$
- E) $Y = A \cdot \overline{B}$
- F) $Y = A \cdot B$

Answer(s):

Mark: 1

2. With regard to MOSFETs, choose all the options which are correct:

- A) A MOSFET may be modelled as current-controlled voltage source.
- B) A MOSFET may be modelled as voltage-controlled current source.
- C) A MOSFET may be modelled as current-controlled current source.
- D) Gate-source voltage (V_{GS}) controls drain-to-source current (I_{DS})
- E) Gate-source voltage (V_{GS}) controls drain-to-gate current (I_{GS})
- F) In an N-channel enhancement mode MOSFET, the channel is formed when V_{GS} is above the threshold voltage V_T .
- G) In an N-channel enhancement mode MOSFET, the channel is formed when V_{GS} is below the threshold voltage V_T .

Answer(s):

Mark: 1

Rough work

5. **Write all correct options** with regard to the relay circuit of question 4:

- A) The diode suppresses development of a large voltage across the relay contacts.
- B) The diode suppresses development of a large POSITIVE voltage at the collector when the relay turns OFF.
- C) The diode suppresses development of a large NEGATIVE voltage at the collector when the relay turns OFF.
- D) The diode suppresses development of a large POSITIVE voltage at the collector when the relay turns ON.
- E) The diode suppresses development of a large NEGATIVE voltage at the collector when the relay turns ON.
- F) The diode is used to reduce the current rating of the inductor.

Answer(s):

Mark: 1

6. Out of the following statements regarding battery operated (BO) motors used in the MS101 lab experiment, write **all the correct options**.

- A) The brush-commutator arrangement is required to produce constant torque.
- B) The brush-commutator arrangement is required to produce unidirectional torque.
- C) The brush-commutator arrangement is required to produce variable torque.
- D) The brush-commutator arrangement is required to produce bidirectional torque.
- E) The stator consists of two permanent magnets and the rotor has a set of coils.
- F) The rotor consists of two permanent magnets and the stator has a set of coils.
- G) Both the stator and rotor have permanent magnets and a set of coils.

Answer(s):

Mark: 1

7. Write **all the correct options**. In comparison to brushed DC motors, the brushless DC motors have:

- A) Lower maintenance and longer life-span.
- B) Lower overall construction and material cost.
- C) Higher efficiency.
- D) Lower reliability.
- E) Lower power-to-size ratio.
- F) Lower electromagnetic noise.

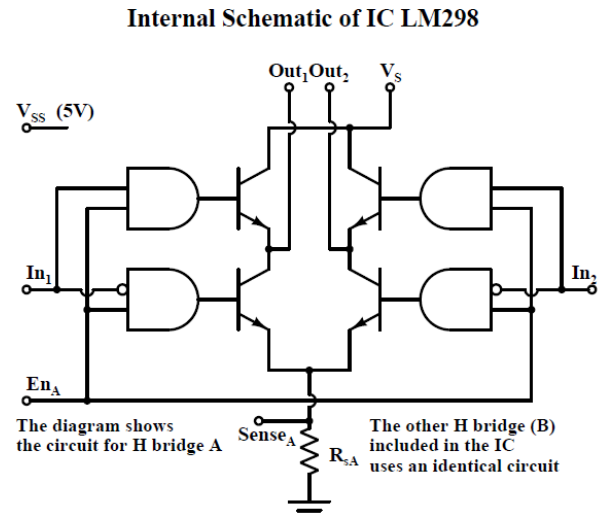
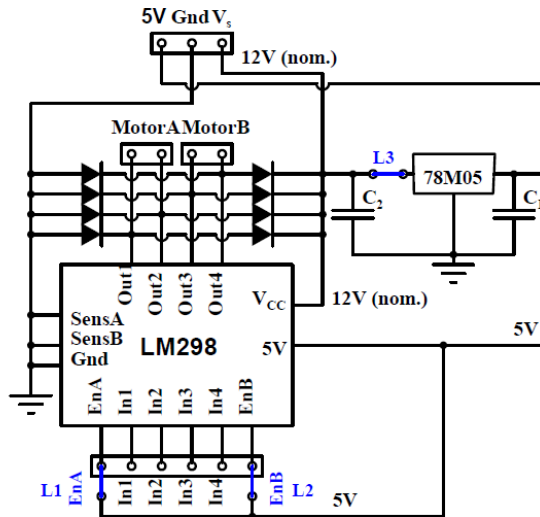
Answer(s):

Mark: 1

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Rough work:

8. The circuit diagrams for the motor driver board L298N and the internal circuit of the IC LM298 used by it are given below.



What is the function of the eight diodes in the board circuit? Write **all correct options**.

- A) To make the motor current flow in the correct direction.
- B) To control the direction of motor rotation (clockwise or counter-clockwise).
- C) To ensure that the pull-up and pull-down transistors are not on simultaneously.
- D) To suppress the noise caused by brush contacts to the inductive load presented by the motor coils.
- E) To protect the output transistors that are driving the motor current.

Answer(s):

Mark: 1

9. Refer to the circuit diagrams for the motor driver board L298N and the internal circuit of the IC LM298 shown in question 8.

For controlling the speed as well as the direction of rotation of the motor marked as MotorA, we should:

(Write all correct options)

- A) Remove Link L1, provide a PWM input to EnA and drive In1 and In2 to '01' or '10'.
- B) Remove Link L1, provide a PWM input to EnA and drive In1 and In2 to '00' or '11'.
- C) Keep Link L1 in place and drive In1 and In2 with '01' or '10'.
- D) Keep Link L1 in place and drive In1 and In2 with '00' or '11'.

Answer(s):

Mark: 1

Rough work:

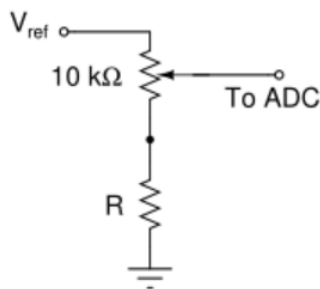
Part B: EE Project Quiz, Marks 10

Q.no.	1	2	3	4	5	6					TOTAL
Marks											

1. In a motor speed controller, we input a PWM signal to the motor driver to control the motor speed. The PWM signal is generated by a micro-controller card which uses the output of an internal ADC to control the duty cycle of the PWM signal. The output of the ADC is an 8-bit number n , which is proportional to an analog voltage input provided by a potential divider using a $10\text{ k}\Omega$ potentiometer and a fixed resistor R as shown in the figure below.

It is observed that the motor does not rotate at all for $n < 55$. When the slider of the potentiometer is at the top, V_{ref} is applied to the ADC, which produces $n = 255$.

What should be the value of R , such that $n = 55$ when the slider of the potentiometer is at the bottom.



Marks: 3

2. Consider a light interrupter disk with 8 dark and 8 light sectors as shown in the figure below. This disk is rotated using a motor and interrupts a beam of light which is sensed by a light sensor. The output of the sensor is viewed on an oscilloscope and is found to be a trapezoidal wave shape with a period of 62.5 ms . What is rotation speed of the motor in revolutions per minute (RPM)?



Marks: 2

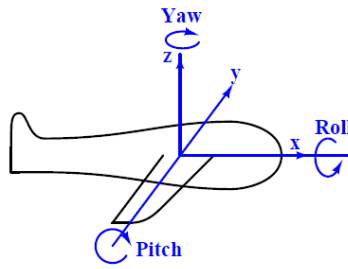


Fig. An object (drone) flying along the x direction (reference figure for questions 3 to 6)

3. While controlling a drone in flight, measurement of the tilt of the drone requires values of:
Write **all correct options**.

- A) The acceleration due to gravity (g) along the 3 axes in the drone's frame of reference.
- B) Value of g as well as angular momentum around the 3 axes.
- C) Value of g along the vertical axis of the drone and the angular momentum around it.
- D) Only the value of g along the vertical axis of the drone.

Answer(s):

Mark: 1

4. Estimation of the restoring force required to bring a tilted drone to a horizontal position requires:
Write **all correct options**.

- A) The acceleration due to gravity (g) along the 3 axes in the drone's frame of reference.
- B) Value of g as well as angular momentum around the 3 axes.
- C) Value of g along the vertical axis of the drone and the angular momentum around it.
- D) Only the value of g along the vertical axis of the drone.

Answer(s):

Mark: 1

5. Incorrect pitch in a drone is corrected by
Write **all correct options**.

- A) applying differential speed inputs to the 2 left propellers and 2 right propellers.
- B) applying differential speed inputs to the 2 front propellers and 2 back propellers.
- C) applying differential speed inputs to the one left propeller and one front propeller.
- D) applying differential speed inputs to the one right propeller and one back propeller.
- E) increasing the throttle.
- F) decreasing the throttle.

Answer(s):

Mark: 1

6. Incorrect roll is corrected by
Write **all correct options**

- A) applying differential speed inputs to the 2 left propellers and 2 right propellers.
- B) applying differential speed inputs to the 2 front propellers and 2 back propellers.
- C) applying differential speed inputs to the one left propeller and one front propeller.
- D) applying differential speed inputs to the one right propeller and one back propeller.
- E) increasing the throttle.
- F) decreasing the throttle.

Answer(s):

Mark: 1